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Karl Deisseroth

Karl Alexander Deisseroth (born November 18, 1971) is an American scientist and physician. He is the D.H. Chen Foundation Professor of Bioengineering and a professor of psychiatry and behavioral sciences at Stanford University. He was jointly awarded the Nobel Prize in Physiology or Medicine in 2026 for his work in optogenetics.[2]

He is known for developing the technologies of hydrogel-tissue chemistry (e.g., CLARITY and STARmap) and optogenetics, and for applying integrated optical and genetic strategies to study normal neural circuit function and dysfunction in neurological and psychiatric illnesses.

In 2019, Deisseroth was elected a member of the United States National Academy of Engineering for his discoveries and control of neuronal signals underlying animal behavior in health and disease, using molecular and optical tools. He is also a member of the United States National Academy of Sciences and the US National Academy of Medicine. In 2026, he was jointly awarded the Nobel Prize in Physiology or Medicine with Georg Nagel and Peter Hegemann "for their discoveries concerning light-gated ion channels and optogenetics".[2]

Early life and education

Karl Deisseroth was born in Boston and later lived in Houston, graduating from St. John's School in 1988.[3] He attended Harvard University and earned a Bachelor of Arts in biochemical sciences in 1992. He moved to Stanford and earned a PhD in neuroscience from Stanford University in 1998, followed by an MD in 2000.[4] He completed his medical internship and psychiatry residency at Stanford University School of Medicine.[5][6]

Academic career

Deisseroth has led his laboratory at Stanford University since 2004. He serves as an attending physician at Stanford Hospital and Clinics and has been affiliated with the Howard Hughes Medical Institute (HHMI) since 2009.[7][8] Between 2014 and 2019, he was a foreign adjunct professor at Sweden's Karolinska Institute.[9]

In 2021, he authored a book titled *Projections: A Story of Human Emotions*, published by Random House, in which he explored the origins of human emotions through personal encounters with neuropsychiatric patients.[10]

Academic research

Optogenetics

Deisseroth named this field "optogenetics" in 2006 and followed up with the development of optogenetic technologies, leading to many applications, including in psychiatry and neurology. In 2010, the journal *Nature Methods* named optogenetics "Method of the Year".[11]

For developing optogenetics, Deisseroth received in 2010 the Nakasone Award; in 2013 the Lounsbery Award and the Dickson Prize in Science; in 2014 the Keio Medical Science Prize; and in 2015 the Albany Prize, Lurie Prize, Dickson Prize in Medicine, and Breakthrough Prize in Life Sciences.[12] He also received the 2015 BBVA Foundation Frontiers of Knowledge Award in Biomedicine, jointly with Edward Boyden and Gero Miesenböck. In 2016, Deisseroth received the Massry Prize along with Peter Hegemann and Miesenböck for "optogenetics, a technology that utilizes light to control cells in living tissues".[13] In 2016, Deisseroth and Hegemann received the Harvey Prize from Technion in Israel "for their discovery of opsin molecules, involved in sensing light in microorganisms, and their pioneering work in using these opsins to develop optogenetics".[14] Deisseroth was then awarded Japan's highest private prize, the Kyoto Prize, in 2018, for "his discovery of optogenetics and the development of causal systems neuroscience", becoming the youngest recipient of the award to date.[15][16] In 2019, Deisseroth, Hegemann, Boyden, and Miesenböck won the Warren Alpert Foundation Prize.[17] Finally in 2020, Deisseroth received the Heineken Prize from the Royal Netherlands Academy of Arts and Sciences, "for developing optogenetics — a method to influence the activity of nerve cells with light".[18]

Deisseroth is also known for achieving insight into the light-gated ion channel pore of channelrhodopsin itself, through his team's initial high-resolution crystal structures of cation and anion-conducting channelrhodopsins[19][20][21] and through a body of structure/function work discovering mechanisms of channelrhodopsin kinetics, ion selectivity, and color selectivity, together with his frequent collaborator Peter Hegemann.[22]

Although the first peer-reviewed paper[23] demonstrating activation of neurons with a channelrhodopsin was from his lab in mid-2005, Deisseroth has emphasized that many "pioneering laboratories around the world"[24] were also working on the idea and published their papers within the following year; he cites Stefan Herlitze[25] and Alexander Gottschalk/Georg Nagel,[26] who published their papers in late 2005, and Hiromu Yawo[27] and Zhuo-Hua Pan,[28] who published their initial papers in 2006 (Pan's early observation of optical activation of retinal neurons expressing channelrhodopsin would have occurred in August 2004, according to Pan,[29] about a month after Deisseroth's initial observation). Deisseroth has published the notebook pages from early July 2004 of his initial experiment showing light activation of neurons expressing a channelrhodopsin.[24] Deisseroth also pointed out[24] that an even earlier experiment had occurred and was published by Heberle and Büldt in 1994, in which functional heterologous expression of a bacteriorhodopsin for light-activated ion flow had been published in a non-neural system (yeast).[30] Optogenetics with microbial opsins as a general technology for neuroscience was enabled only by the full development of versatile strategies for targeting opsins and light to specific cells in behaving animals by taking advantage of Cre-lox neurogenetics developed by Joe Tsien in the 1990s.[24][31][32]

Karl Deisseroth



Deisseroth in 2025

Born	November 18, 1971 Boston, Massachusetts, U.S.[1]
Education	Harvard University (BA) Stanford University (MD–PhD)
Known for	Optogenetics and hydrogel-tissue chemistry (including CLARITY and STARmap)
Spouse	Michelle Monje
Awards	W. Alden Spencer Award (2011) Keio Medical Science Prize (2014) Albany Medical Center Prize (2015) BBVA Foundation Frontiers of Knowledge Award (2015) Breakthrough Prize in Life Sciences (2016) Kyoto Prize (2018) Heineken Prize (2020) Albert Lasker Award for Basic Medical Research (2021) Louisa Gross Horwitz Prize (2022) Japan Prize (2023) Nobel Prize in Physiology or Medicine (2026)
Scientific career	
Fields	Neuroscience Psychiatry Bioengineering
Workplaces	Stanford University
Thesis	*Communication between Synapse and Nucleus (https://search.worldcat.org/title/81696896)* (1998)
Doctoral advisor	Richard Tsien
Other academic advisors	Robert Malenka
Doctoral students	Feng Zhang Viviana Gradinaru

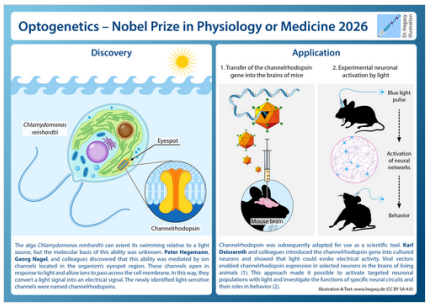
Functional materials

Deisseroth is also known for a separate class of technological innovation. His group has developed methods for chemically assembling functional materials within biological tissue. This approach has a range of applications, including probing the molecular composition and wiring of cells within intact brains.

In 2013, Deisseroth was the senior author of a paper describing the initial form of this method, called CLARITY (with a team including first author Kwanghun Chung, a postdoctoral fellow in his lab,[33] and neuroscientist Viviana Gradinaru).[34] This method makes biological tissues, such as mammalian brains, translucent and accessible to molecular probes.[35] CLARITY[36] has been widely used,[37] and many variants on the basic HTC backbone have been developed in other labs as well since 2013 (reviewed in[38]).

A key feature of HTC is that the hydrogel-tissue hybrid "becomes the substrate for future chemical and optical interrogation that can be probed and manipulated in new ways".[38] For example, HTC variants now enable improved anchoring and amplification of RNA, reversible size changes (contraction or expansion), and in situ sequencing (reviewed in[38]). In particular, STARmap is an HTC variant that allows three-dimensional cellular-resolution transcriptomic readouts within intact tissue.[39][40][41])

Other notable students	Edward Boyden Kay Tye Xiao Wang Priya Rajasethupathy Ilana B. Witten
Website	deisseroth.com (https://deisseroth.com)



Nobel Prize 2026 for the development of optogenetics

Personal life

Deisseroth is married to neuroscientist Michelle Monje, with whom he has four children.[42][43][44][45] He also has a son from a previous relationship.[46]

Lists

List of selected publications

Deisseroth has published influential work on optogenetics, neural-circuit methods, and tissue-clearing techniques, while also writing about neuroscience for general audiences.[47][48][49]

Academic

- Boyden, E. S.; Zhang, F.; Bamberg, E.; Nagel, G.; Deisseroth, K. (2005). "Millisecond-timescale, genetically targeted optical control of neural activity." *Nature Neuroscience*. 8: 1263–1268.
- Chung, K.; Deisseroth, K. (2013). "CLARITY for mapping the nervous system." *Nature Methods*. 10: 508–513.
- Deisseroth, K. (2015). "Optogenetics: 10 years of microbial opsins in neuroscience." *Nature Neuroscience*. 18: 1213–1225.

Public writing

- Deisseroth, K. (2010). "Controlling the Brain with Light." *Scientific American*.
- Deisseroth, K. (2016). "Building a See-Through Brain." *Scientific American*.

List of honors and awards

Two major prizes paid particular attention to Deisseroth's work on elucidation of the structure and function of light-gated ion channels—the 2016 Harvey Prize to Deisseroth and Hegemann for the "discovery of opsin molecules, involved in sensing light in microorganisms, and for the pioneering work in utilizing these opsins to develop optogenetics",[14] and the 2018 Gairdner Award, which noted "his group discovered the fundamental principles of the unique channelrhodopsin proteins in molecular detail by a wide range of genomic, biophysical, electrophysiological and structural techniques with many mutants in close collaboration with Peter Hegemann").[50] Several major prizes have cited Deisseroth's development of HTC, including:

- The 2017 Fresenius Prize "for his discoveries in optogenetics and hydrogel-tissue chemistry, as well as his research into the neural circuit basis of depression".[51][52]
- The 2015 Lurie Prize in Biomedical Sciences "for leading the development of optogenetics, a technology for controlling cells with light to determine function, as well as for CLARITY, a method for transforming intact organs into transparent polymer gels to allow visualization of biological structures with high resolution and detail".[53]
- The 2013 Premio Citta di Firenze[54]
- The Redelsheimer Award for "optogenetics, CLARITY, and other novel and powerful neural circuit approaches in furthering the field's understanding of the neuroscience underlying behavior".[55]
- The 2015 Dickson Prize in Medicine[56]
- The 2020 Heineken Prize for Medicine, for "developing optogenetics — a method to influence the activity of nerve cells with light — as well as for developing hydrogel-tissue chemistry, which enables researchers to make biological tissue accessible to light and molecular probes."[57]
- Deisseroth's 2018 Kyoto Prize cited his "causal systems neuroscience".[58]
- The 2013 Pasarow Prize[59] was awarded to Deisseroth for "neuropsychiatry research".
- The 2013 Premio Citta di Firenze was given to Deisseroth for "innovative technologies to probe the structure and dynamics of circuits related to schizophrenia, autism, narcolepsy, Parkinson's disease, depression, anxiety and addiction".[54]
- The Redelsheimer Award from the Society for Biological Psychiatry was awarded to Deisseroth for "furthering the field's understanding of the neuroscience underlying behavior".[60]
- Deisseroth's 2017 Fresenius Prize cited "his discoveries in optogenetics and hydrogel-tissue chemistry, as well as his research into the neural circuit basis of depression".[61]
- 2005 Presidential Early Career Award for Scientists and Engineers[62]
- 2010 Nakasone Award, Human Frontier Science Program[63]
- 2010 Koetser Award for Brain Research[64]
- 2011 W. Alden Spencer Award[65]
- 2012 Zuelch Prize, with Peter Hegemann, Georg Nagel, and Ernst Bamberg[66]
- 2012 Perl-UNC Prize[67]
- 2013 Premio Città di Firenze[68]
- 2013 Goldman-Rakic award, Brain & Behavior Research Foundation[69]
- 2013 Jacob Heskel Gabbay Award[70]
- 2013 Brain Prize[71]
- 2013 Robert J. and Claire Pasarow Foundation Medical Research Award
- 2013 Richard Lounsbery Award[72]

- 2013 Dickson Prize in Science[73]
- 2014 Keio Medical Science Prize[74]
- 2015 Albany Medical Center Prize[75]
- 2015 Lurie Prize in Biomedical Sciences[76]
- 2015 Breakthrough Prize in Life Sciences[12]
- 2015 Dickson Prize in Medicine[77]
- 2015 BBVA Foundation Frontiers of Knowledge Award[78]
- 2016 Massry Prize, with Peter Hegemann and Gero Miesenböck[79]
- 2017 Redelsheimer Award, Society for Biological Psychiatry[80]
- 2017 Fresenius Prize, Else Kröner-Fresenius Foundation[81]
- 2017 Harvey Prize, with Peter Hegemann[82]
- 2018 Leibinger Prize[83]
- 2018 Eisenberg Prize, University of Michigan[84]
- 2018 Canada Gairdner International Award[85]
- 2018 Kyoto Prize (Advanced Technology)[86]
- 2019 Warren Alpert Foundation Prize, with Ed Boyden, Peter Hegemann, and Gero Miesenböck[17]
- 2019 National Academy of Engineering membership
- 2020 Heineken Prize for Medicine[57]
- 2021 Albert Lasker Award for Basic Medical Research[87]
- 2022 Louisa Gross Horwitz Prize[88]
- 2023 Japan Prize, with Gero Miesenböck[89]
- 2025 Asan Award in Medicine[90][91]
- 2026 Nobel Prize in Physiology or Medicine[2]

References

1. "Karl Deisseroth Facts" (https://www.nobelprize.org/prizes/medicine/2026/deisseroth/facts/). *The Nobel Prize*. October 5, 2026. Retrieved October 5, 2026.
2. "Nobel Prize in Physiology or Medicine 2026" (https://www.nobelprize.org/prizes/medicine/2026/press-release/). *NobelPrize.org*. Archived (https://web.archive.org/web/20261005094452/https://www.nobelprize.org/prizes/medicine/2026/press-release/) from the original on October 5, 2026. Retrieved October 5, 2026.
3. "Karl Deisseroth Facts" (https://www.nobelprize.org/prizes/medicine/2026/deisseroth/facts/). *The Nobel Prize*. October 5, 2026. Retrieved October 5, 2026.
4. "The Nobel Prize in Physiology or Medicine 2026 – Press release" (https://www.nobelprize.org/prizes/medicine/2026/press-release/). *NobelPrize.org*. October 5, 2026. Retrieved October 10, 2026.
5. "Karl Deisseroth | Stanford Medicine" (https://med.stanford.edu/profiles/karl-deisseroth). *CAP Profiles*. Archived (https://web.archive.org/web/20250913145137/https://med.stanford.edu/profiles/karl-deisseroth) from the original on September 13, 2025. Retrieved August 11, 2025.
6. "Karl Deisseroth – NAS" (https://www.nasonline.org/directory-entry/karl-deisseroth-4gxnxv/). *nasonline.org*. Retrieved August 11, 2025.
7. "Karl Deisseroth, MD, PhD" (http://www.hhmi.org/scientists/karl-deisseroth). *HHMI.org*. Archived (https://web.archive.org/web/20190422055427/https://www.hhmi.org/scientists/karl-deisseroth) from the original on April 22, 2019. Retrieved March 1, 2016.
8. Smith, Kerri (May 29, 2013). "Neuroscience: Method man" (http://www.nature.com/news/neuroscience-method-man-1.13077). *Nature News*. Archived (https://web.archive.org/web/20170917070956/http://www.nature.com/news/neuroscience-method-man-1.13077) from the original on September 17, 2017. Retrieved February 27, 2014.
9. "Karl Deisseroth, M.D., Ph.D." (https://embc.embs.org/2016/karl-deisseroth/) *EMBC 2016*. January 20, 2018. Archived (https://web.archive.org/web/20260418134049/https://embc.embs.org/2016/karl-deisseroth/) from the original on April 18, 2026. Retrieved October 5, 2026.
10. Deisseroth, Karl (2021). *Projections* (https://www.penguinrandomhouse.com/books/600209/projections-by-karl-deisseroth/). New York: Random House. ISBN 978-1-9848-5369-1. Archived (https://web.archive.org/web/20220118120111/https://www.penguinrandomhouse.com/books/600209/projections-by-karl-deisseroth/) from the original on January 18, 2022. Retrieved January 18, 2022.
11. "Method of the Year 2010" (https://doi.org/10.1038%2Fnmeth.f.321). *Nature Methods*. **8** (1): 1. December 20, 2010. doi:10.1038/nmeth.f.321 (https://doi.org/10.1038%2Fnmeth.f.321).
12. "Breakthrough Prize" (https://breakthroughprize.org/Laureates/2). Breakthrough Prize. Archived (https://web.archive.org/web/20181212163208/https://breakthroughprize.org/Laureates/2) from the original on December 12, 2018. Retrieved March 1, 2016.
13. "MASSRY PRIZE 2016" (https://keck.usc.edu/massry-prize/wp-content/uploads/sites/168/2016/06/Flyer_Massry-2016.pdf) (PDF). *keck.usc.edu*. Archived (https://web.archive.org/web/20190129232800/https://keck.usc.edu/massry-prize/wp-content/uploads/sites/168/2016/06/Flyer_Massry-2016.pdf) (PDF) from the original on January 29, 2019. Retrieved September 15, 2020.
14. "Harvey Prize – Prize Winners" (https://harveypz.net.technion.ac.il/harvey-prize-laureates/). Archived (https://web.archive.org/web/20221205084658/https://harveypz.net.technion.ac.il/harvey-prize-laureates/) from the original on December 5, 2022. Retrieved September 15, 2020.
15. "2018 Kyoto Prize Laureates received their Awards on November 10 | Kyoto Prize USA" (https://web.archive.org/web/20190322023829/http://kyotoprize-us.org/announcement-of-the-2018-kyoto-prize-laureates/). Archived from the original (http://kyotoprize-us.org/announcement-of-the-2018-kyoto-prize-laureates/) on March 22, 2019. Retrieved June 12, 2019.
16. Scovie, Jay (2018). "Stanford Neuroscientist Karl Deisseroth Receives Japan's Kyoto Prize in Advanced Technology" (https://www.businesswire.com/news/home/20181117005019/en/Stanford-Neuroscientist-Karl-Deisseroth-Receives-Japan%E2%80%99s-Kyoto). *Business Wire*. Archived (https://web.archive.org/web/20181118073746/https://www.businesswire.com/news/home/20181117005019/en/Stanford-Neuroscientist-Karl-Deisseroth-Receives-Japan%E2%80%99s-Kyoto) from the original on November 18, 2018. Retrieved June 13, 2019.
17. "2019 Warren Alpert Prize Recipients Announced | Warren Alpert Foundation Prize" (https://web.archive.org/web/20210814231223/https://warrenalpert.org/news/2019-warren-alpert-prize-recipients-announced). *warrenalpert.org*. Archived from the original (https://warrenalpert.org/news/2019-warren-alpert-prize-recipients-announced) on August 14, 2021. Retrieved July 16, 2019.
18. "Karl Deisseroth wins 2020 Heineken Prize for Medicine" (http://med.stanford.edu/news/all-news/2020/06/karl-deisseroth-wins-2020-heineken-prize-for-medicine.html). *News Center*. June 3, 2020. Archived (https://web.archive.org/web/20220723201754/https://med.stanford.edu/news/all-news/2020/06/karl-deisseroth-wins-2020-heineken-prize-for-medicine.html) from the original on July 23, 2022. Retrieved July 23, 2022.
19. Kim YS (September 2018). "Crystal Structure of the Natural Anion-Conducting Channelrhodopsin GtACR1" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6340299). *Nature*. **561** (7723): 343–348. Bibcode:2018Natur.561..343K (https://ui.adsabs.harvard.edu/abs/2018Natur.561..343K). doi:10.1038/s41586-018-0511-6 (https://doi.org/10.1038%2Fs41586-018-0511-6). PMC 6340299 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6340299). PMID 30158696 (https://pubmed.ncbi.nlm.nih.gov/30158696).
20. Kato HE (January 22, 2012). "Crystal Structure of the Channelrhodopsin Light-Gated Cation Channel" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4160518). *Nature*. **365** (6453): 369–74. Bibcode:2012Natur.482..369K (https://ui.adsabs.harvard.edu/abs/2012Natur.482..369K). doi:10.1038/nature10870 (https://doi.org/10.1038%2Fnature10870). PMC 4160518 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4160518). PMID 22266941 (https://pubmed.ncbi.nlm.nih.gov/22266941).
21. Kato HE (September 2018). "Structural mechanisms of selectivity and gating in anion channelrhodopsins" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6317992). *Nature*. **561** (7723): 349–354. Bibcode:2018Natur.561..349K (https://ui.adsabs.harvard.edu/abs/2018Natur.561..349K). doi:10.1038/s41586-018-0504-5 (https://doi.org/10.1038%2Fs41586-018-0504-5). PMC 6317992 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6317992). PMID 30158697 (https://pubmed.ncbi.nlm.nih.gov/30158697).
22. Deisseroth K; Hegemann P (September 15, 2017). "The Form and Function of Channelrhodopsin" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5723383). *Science*. **357** (6356) eaan5544. Bibcode:2017Sci...35765544D (https://ui.adsabs.harvard.edu/abs/2017Sci...35765544D). doi:10.1126/science.aan5544 (https://doi.org/10.1126%2Fscience.aan5544). PMC 5723383 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5723383). PMID 28912215 (https://pubmed.ncbi.nlm.nih.gov/28912215).
23. Boyden ES; Zhang F; Bamberg E; Nagel G; Deisseroth K (September 2005). "Millisecond-timescale, genetically targeted optical control of neural activity". *Nature Neuroscience*. **8** (9): 1263–8. doi:10.1038/nn1525 (https://doi.org/10.1038%2Fnn1525). PMID 16116447 (https://pubmed.ncbi.nlm.nih.gov/16116447). S2CID 6809511 (https://api.semanticscholar.org/CorpusID:6809511).
24. Deisseroth K. (September 2015). "Optogenetics: 10 years of microbial opsins in neuroscience" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4790845). *Nature Neuroscience*. **18** (9): 1213–25. doi:10.1038/nn.4091 (https://doi.org/10.1038%2Fnn.4091). PMC 4790845 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4790845). PMID 26308982 (https://pubmed.ncbi.nlm.nih.gov/26308982).
25. Li X; Gutierrez DV; Hanson MG; Han J; Mark MD; Chiel H; Hegemann P; Landmesser LT; Herlitze S (December 6, 2005). "Fast Noninvasive Activation and Inhibition of Neural and Network Activity by Vertebrate Rhodopsin and Green Algae Channelrhodopsin" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1292990). *Proc Natl Acad Sci USA*. **102** (49): 17816–21. Bibcode:2005PNAS..10217816L (https://ui.adsabs.harvard.edu/abs/2005PNAS..10217816L). doi:10.1073/pnas.0509030102 (https://doi.org/10.1073%2Fpnas.0509030102). PMC 1292990 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1292990). PMID 16306259 (https://pubmed.ncbi.nlm.nih.gov/16306259).
26. Nagel G; Brauner M; Liewald J; Adeishvili N; Bamberg E; Gottschalk A (December 2005). "Light activation of channelrhodopsin-2 in excitable cells of Caenorhabditis elegans triggers rapid behavioral responses" (https://doi.org/10.1016%2Fj.cub.2005.11.032). *Current Biology*. **15** (24): 2279–2284. Bibcode:2005CBio...15.2279N (https://ui.adsabs.harvard.edu/abs/2005CBio...15.2279N). doi:10.1016/j.cub.2005.11.032 (https://doi.org/10.1016%2Fj.cub.2005.11.032). PMID 16360690 (https://pubmed.ncbi.nlm.nih.gov/16360690). S2CID 7036529 (https://api.semanticscholar.org/CorpusID:7036529).

27. Ishizuka T; Kakuda M; Araki R; Yawo H (2006). "Kinetic Evaluation of Photosensitivity in Genetically Engineered Neurons Expressing Green Algae Light-Gated Channels". *Neuroscience Research*. **54** (2): 85–94. doi:10.1016/j.neures.2005.10.009 (https://doi.org/10.1016%2Fj.neures.2005.10.009). PMID 16298005 (https://pubmed.ncbi.nlm.nih.gov/16298005). S2CID 17576414 (https://api.semanticscholar.org/CorpusID:17576414).
28. Bi A; Cui J; Ma Y; Olshevskaya E; Pu M; Dizhoor A; Pan Z (2006). "Ectopic expression of a microbial-type rhodopsin restores visual responses in mice with photoreceptor degeneration" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1459045). *Neuron*. **50** (1): 23–33. doi:10.1016/j.neuron.2006.02.026 (https://doi.org/10.1016%2Fj.neuron.2006.02.026). PMC 1459045 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1459045). PMID 16600853 (https://pubmed.ncbi.nlm.nih.gov/16600853).
29. "He may be the rightful inventor of neuroscience's biggest breakthrough in decades. But you've never heard of him" (https://www.statnews.com/2016/09/01/optogenetics/). *STAT*. September 1, 2016. Archived (https://web.archive.org/web/20210326075219/https://www.statnews.com/2016/09/01/optogenetics/) from the original on March 26, 2021. Retrieved February 9, 2020.
30. Hoffman A; Hildebrandt V; Heberle J; Buldt G (1994). "Photoactive mitochondria: in vivo transfer of a light-driven proton pump into the inner mitochondrial membrane of Schizosaccharomyces pombe" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC44813). *Proc. Natl. Acad. Sci. USA*. **91** (20): 9367–9371. Bibcode:1994PNAS...91.9367H (https://ui.adsabs.harvard.edu/abs/1994PNAS...91.9367H). doi:10.1073/pnas.91.20.9367 (https://doi.org/10.1073%2Fpnas.91.20.9367). PMC 44813 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC44813). PMID 7937771 (https://pubmed.ncbi.nlm.nih.gov/7937771).
31. Joe Z. Tsien; et al. (1996). "Subregion- and cell type-restricted gene knockout in mouse brain" (https://doi.org/10.1016%2FS0092-8674%2800%2981826-7). *Cell*. **87** (7): 1317–1326. doi:10.1016/S0092-8674(00)81826-7 (https://doi.org/10.1016%2FS0092-8674%2800%2981826-7). PMID 8980237 (https://pubmed.ncbi.nlm.nih.gov/8980237). S2CID 863399 (https://api.semanticscholar.org/CorpusID:863399).
32. Tsien JZ. (2016). Cre-lox neurogenetics: 20 years of versatile applications in brain research and counting...Front. Genet. | doi:10.3389/fgene.2016.00019 (https://doi.org/10.3389%2Ffgene.2016.00019) http://journal.frontiersin.org/article/10.3389/fgene.2016.00019/abstract
33. Deisseroth KA, Chung K. 2015. Methods and compositions for preparing biological specimens for microscopic analysis. www.google.com/patents/US20150144490. Filing date: March 13, 2013. US Patent Appl. No. US20150144490
34. Deisseroth KA, Gradinaru V. 2014. Functional targeted brain endoskeletonization. www.google.com/patents/US20140030192. Filing date: January 26, 2012. US Patent Appl. No. US20140030192.
35. "Brains as Clear as Jell-O for Scientists to Explore" (https://www.nytimes.com/2013/04/11/science/brains-as-clear-as-jell-o-for-scientists-to-explore.html) Archived (https://web.archive.org/web/20171102234843/http://www.nytimes.com/2013/04/11/science/brains-as-clear-as-jell-o-for-scientists-to-explore.html) November 2, 2017, at the Wayback Machine, April 10, 2013 *The New York Times*
36. "CLARITY Wiki" (http://wiki.claritytechniques.org/index.php/Main_Page). *wiki.claritytechniques.org*.
37. "Journal Articles – CLARITY Wiki" (http://wiki.claritytechniques.org/index.php/Journal_Articles). *wiki.claritytechniques.org*. Archived (https://web.archive.org/web/20200226003053/http://wiki.claritytechniques.org/index.php/Journal_Articles) from the original on February 26, 2020. Retrieved April 19, 2020.
38. Viviana Gradinaru; Jennifer Treweek; Kristin Overton; Karl Deisseroth (2018). "Hydrogel-tissue chemistry: principles and applications" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6359929). *Annual Review of Biophysics*. **47**: 355–376. doi:10.1146/annurev-biophys-070317-032905 (https://doi.org/10.1146%2Fannurev-biophys-070317-032905). PMC 6359929 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6359929). PMID 29792820 (https://pubmed.ncbi.nlm.nih.gov/29792820).
39. Wang, X (June 21, 2018). "Three-dimensional intact-tissue sequencing of single-cell transcriptional states" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6339868). *Science*. **361** (6400). doi:10.1126/science.361.6400.375-l (https://doi.org/10.1126%2Fscience.361.6400.375-l). PMC 6339868 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6339868). PMID 29930089 (https://pubmed.ncbi.nlm.nih.gov/29930089).
40. Thomas Knopfel (July 27, 2018). "Neurotechnology to address big questions". *Science*. **361** (6400): 328–329. Bibcode:2018Sci...361..328K (https://ui.adsabs.harvard.edu/abs/2018Sci...361..328K). doi:10.1126/science.aau4705 (https://doi.org/10.1126%2Fscience.aau4705). hdl:10044/1/71425 (https://hdl.handle.net/10044%2F1%2F71425). PMID 30049862 (https://pubmed.ncbi.nlm.nih.gov/30049862). S2CID 50787948 (https://api.semanticscholar.org/CorpusID:50787948).
41. "STARmap Resources" (https://web.archive.org/web/20210503082015/https://www.starmapresources.com/). *STARmap Resources*. Archived from the original (https://www.starmapresources.com/) on May 3, 2021. Retrieved May 3, 2021.
42. "Michelle Monje" (https://brainprize.org/winners/cancer-neuroscience-2025/michelle-monje). *The Brain Prize*. Archived (https://web.archive.org/web/20250307161033/https://brainprize.org/winners/cancer-neuroscience-2025/michelle-monje) from the original on March 7, 2025. Retrieved August 11, 2025.
43. "Love means sharing the champagne" (https://news.stanford.edu/stories/2022/02/love-means-sharing-champagne). *news.stanford.edu*. Archived (https://web.archive.org/web/20250520190927/https://news.stanford.edu/stories/2022/02/love-means-sharing-champagne) from the original on May 20, 2025. Retrieved August 11, 2025.
44. Godwin, Richard (June 12, 2021). "Neuroscientist Karl Deisseroth: 'Coronavirus has changed us all' " (https://www.theguardian.com/books/2021/jun/12/neuroscientist-karl-deisseroth-coronavirus-has-changed-us-all). *The Guardian*. Retrieved August 11, 2025.
45. KRON 4 (October 5, 2026). *Stanford professor wins Nobel Prize* (https://www.youtube.com/watch?v=KKZY3vErtvw). Retrieved October 6, 2026 – via YouTube.
46. "Love means sharing the champagne" (https://news.stanford.edu/stories/2022/02/love-means-sharing-champagne). *news.stanford.edu*. Archived (https://web.archive.org/web/20250520190927/https://news.stanford.edu/stories/2022/02/love-means-sharing-champagne) from the original on May 20, 2025. Retrieved August 11, 2025. "*Michelle: And then the oldest, who is my stepson, is 25 now. He's an MD-PhD student at Baylor.*"
47. Boyden, Edward S; Zhang, Feng; Bamberg, Ernst; Nagel, Georg; Deisseroth, Karl (September 2005). "Millisecond-timescale, genetically targeted optical control of neural activity" (https://www.nature.com/articles/nn1525). *Nature Neuroscience*. **8** (9): 1263–1268. doi:10.1038/nn1525 (https://doi.org/10.1038%2Fnn1525). ISSN 1097-6256 (https://search.worldcat.org/issn/1097-6256). PMID 16116447 (https://pubmed.ncbi.nlm.nih.gov/16116447).
48. Chung, Kwanghun; Deisseroth, Karl (June 2013). "CLARITY for mapping the nervous system" (https://www.nature.com/articles/nmeth.2481). *Nature Methods*. **10** (6): 508–513. doi:10.1038/nmeth.2481 (https://doi.org/10.1038%2Fnmeth.2481). ISSN 1548-7091 (https://search.worldcat.org/issn/1548-7091). PMID 23722210 (https://pubmed.ncbi.nlm.nih.gov/23722210).
49. Deisseroth, Karl (September 2015). "Optogenetics: 10 years of microbial opsins in neuroscience" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4790845). *Nature Neuroscience*. **18** (9): 1213–1225. doi:10.1038/nn.4091 (https://doi.org/10.1038%2Fnn.4091). ISSN 1097-6256 (https://search.worldcat.org/issn/1097-6256). PMC 4790845 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4790845). PMID 26308982 (https://pubmed.ncbi.nlm.nih.gov/26308982).
50. "Karl Deisseroth – 2018 Gairdner Awards Laureate" (https://gairdner.org/award_winners/karl-deisseroth/). Archived (https://web.archive.org/web/20211130161353/https://gairdner.org/award_winners/karl-deisseroth/) from the original on November 30, 2021. Retrieved December 13, 2021.
51. "Else Kröner Fresenius Prize for Medical Research 2017 | Else Kröner-Fresenius-Stiftung" (https://www.ekfs.de/en/scientific-funding/international-research-prize/else-kroener-fresenius-prize-for-medical-research-2017). *ekfs.de*.
52. Deisseroth K (2017). "Optical and chemical discoveries recognized for impact on biology and psychiatry" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5452044). *EMBO Reports*. **18** (6): 859–60. doi:10.15252/embr.201744405 (https://doi.org/10.15252%2Fembr.201744405). PMC 5452044 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5452044). PMID 28566521 (https://pubmed.ncbi.nlm.nih.gov/28566521).
53. "FNIH Awards Lurie Prize to Karl Deisseroth | The Foundation for the National Institutes of Health" (https://fnih.org/news/press-releases/fnih-awards-lurie-prize-karl-deisseroth). *fnih.org*. April 27, 2023. Archived (https://web.archive.org/web/20210503082014/https://fnih.org/news/press-releases/fnih-awards-lurie-prize-karl-deisseroth) from the original on May 3, 2021. Retrieved May 3, 2021.
54. "Premio Città di Firenze" (https://www.cerm.unifi.it/premio-citta-di-firenze).
55. "Karl Deisseroth receives Inaugural SOBP Distinguished Redelsheimer Award" (https://web.archive.org/web/20190916213756/https://www.eurekalert.org/pub_releases/2017-05/pmg-kdr042817.php). Archived from the original (https://www.eurekalert.org/pub_releases/2017-05/pmg-kdr042817.php) on September 16, 2019. Retrieved April 2, 2020.
56. "2015 Dickson Prize Winner" (https://web.archive.org/web/20190815125953/http://www.dicksonprize.pitt.edu/recipients/2015-deisseroth.php). Archived from the original (http://www.dicksonprize.pitt.edu/recipients/2015-deisseroth.php) on August 15, 2019. Retrieved April 2, 2020.
57. "Dr A.H. Heineken Prize for Medicine 2020 awarded to Karl Deisseroth — KNAW" (https://web.archive.org/web/20211108014610/https://knaw.nl/en/news/news/dr-a-h-heineken-prize-for-medicine-2020-awarded-to-karl-deisseroth). *knaw.nl*. Archived from the original (https://knaw.nl/en/news/news/dr-a-h-heineken-prize-for-medicine-2020-awarded-to-karl-deisseroth) on November 8, 2021. Retrieved June 12, 2020.
58. "Kyoto Prize, Inamori Foundation" (http://www.kyotoprize.org/en/). *Kyoto Prize, Inamori Foundation*. Archived (https://web.archive.org/web/20191012071611/https://www.kyotoprize.org/en/) from the original on October 12, 2019. Retrieved March 13, 2019.
59. "Three researchers earn Pasarow Awards" (http://med.stanford.edu/news/all-news/2013/03/three-researchers-earn-pasarow-awards.html). *News Center*. March 6, 2013.
60. "Karl Deisseroth receives Inaugural SOBP Distinguished Redelsheimer Award" (https://web.archive.org/web/20190916213756/https://www.eurekalert.org/pub_releases/2017-05/pmg-kdr042817.php). Archived from the original (https://www.eurekalert.org/pub_releases/2017-05/pmg-kdr042817.php) on September 16, 2019. Retrieved April 2, 2020.
61. "Else Kröner Fresenius Prize for Medical Research 2017" (https://www.ekfs.de/en/scientific-funding/prize-for-medical-research/else-kroener-fresenius-prize-for-medical-research-2017). *ekfs.de*. Retrieved September 15, 2020.
62. "White House Announces 2005 Awards for Early Career Scientists and Engineers" (https://web.archive.org/web/20210726131726/http://ftp.csr.utexas.edu/pub/ggfc/misc/PECASE_PR_Release.pdf) (PDF). Office of Science and Technology Policy, Executive Office of the President. July 26, 2006. Archived from the original (http://ftp.csr.utexas.edu/pub/ggfc/misc/PECASE_PR_Release.pdf) (PDF) on July 26, 2021. Retrieved November 12, 2018 – via Center for Space Research, University of Texas at Austin.

63. "Nakasone Award 2010-Karl-Deisseroth" (https://www.hfsp.org/hfsp-nakasone-award/2010-karl-deisseroth). Archived (https://web.archive.org/web/20211102181725/https://www.hfsp.org/hfsp-nakasone-award/2010-karl-deisseroth) from the original on November 2, 2021. Retrieved April 2, 2020.
64. "Koetser Prize Former-Awardees" (https://web.archive.org/web/20220308122759/http://koetserfoundation.org/former-awardees.php). Archived from the original (http://koetserfoundation.org/former-awardees.php) on March 8, 2022. Retrieved April 2, 2020.
65. "34th Annual W. Alden Spencer Award and Lecture" (https://web.archive.org/web/20150427143022/http://calendar.columbia.edu/sundial/webapi/get.php?vt=detail&id=54414&con=standalone&br=cumc). Columbia University. Archived from the original (http://calendar.columbia.edu/sundial/webapi/get.php?vt=detail&id=54414&con=standalone&br=cumc) on April 27, 2015. Retrieved August 24, 2012.
66. "2012 Zuelch Prize" (https://www.mpg.de/1156268/Zuelch_Prize). Archived (https://web.archive.org/web/20180929113239/https://www.mpg.de/1156268/Zuelch_Prize) from the original on September 29, 2018. Retrieved April 2, 2020.
67. "deisseroth-wins-four-awards-seminal-work-optogenetics" (https://biox.stanford.edu/highlight/deisseroth-wins-four-awards-seminal-work-optogenetics). April 21, 2014.
68. "2013 premio-citta-di-firenze" (https://web.archive.org/web/20140617062924/http://www.cerm.unifi.it/news-a-events/premio-citta-di-firenze). Archived from the original (http://www.cerm.unifi.it/news-a-events/premio-citta-di-firenze) on June 17, 2014. Retrieved April 2, 2020.
69. "past-outstanding-achievement-prizewinners#goldman-past" (https://www.bbrfoundation.org/grants-prizes/prizes-awards/past-outstanding-achievement-prizewinners#goldman-past). April 11, 2017.
70. "2013 Gabbay Award Past Winners" (http://www.brandeis.edu/rosenstiel/gabbayaward/past.html). Archived (https://web.archive.org/web/20140719025631/http://www.brandeis.edu/rosenstiel/gabbayaward/past.html) from the original on July 19, 2014. Retrieved September 10, 2016.
71. "Brain Prize Winners" (https://web.archive.org/web/20200513220313/https://www.lundbeckfonden.com/en/thebrainprize/winners/). *Lundbeckfonden* (in Danish). Archived from the original (https://www.lundbeckfonden.com/en/thebrainprize/winners/) on May 13, 2020. Retrieved March 13, 2019.
72. "2013 Lounsbery Award" (https://www8.nationalacademies.org/onpinews/newsitem.aspx?RecordID=01072013a). Archived (https://web.archive.org/web/20130203005250/http://www8.nationalacademies.org/onpinews/newsitem.aspx?RecordID=01072013a) from the original on February 3, 2013. Retrieved April 2, 2020.
73. "Karl Deisseroth, MD, PhD" (https://www.dicksonprize.pitt.edu/past-recipients/karl-deisseroth-md-phd/). *The Dickson Prize in Medicine*. November 18, 1971. Retrieved October 5, 2026.
74. Optogenetics earns Stanford professor Karl Deisseroth the Keio prize in medicine (http://news.stanford.edu/features/2014/optogenetics/) Archived (https://web.archive.org/web/20161016133102/http://news.stanford.edu/features/2014/optogenetics/) October 16, 2016, at the Wayback Machine, Stanford, 2014
75. "Albany Medical College: AlbanyPrize" (https://web.archive.org/web/20181020192628/http://www.amc.edu/academic/AlbanyPrize/index.cfm). *Amc.edu*. August 14, 2015. Archived from the original (http://www.amc.edu/Academic/AlbanyPrize/index.cfm) on October 20, 2018. Retrieved March 1, 2016.
76. "2015 lurie-prize" (https://web.archive.org/web/20150901025834/http://www.fnih.org/work/lurie-prize-biomedical-sciences). Foundation for the National Institutes of Health. Archived from the original (https://fnih.org/what-we-do/programs/lurie-prize) on September 1, 2015. Retrieved August 14, 2015.
77. "Dickson recipients/2015-deisseroth.php" (https://web.archive.org/web/20190815125953/http://www.dicksonprize.pitt.edu/recipients/2015-deisseroth.php). Archived from the original (http://www.dicksonprize.pitt.edu/recipients/2015-deisseroth.php) on August 15, 2019. Retrieved April 2, 2020.
78. "2015 BBVA Award" (https://www.youtube.com/watch?v=e3_J-2MB-FM). *YouTube*. June 24, 2016. Archived (https://web.archive.org/web/20250123114416/https://www.youtube.com/watch?v=e3_J-2MB-FM) from the original on January 23, 2025. Retrieved April 2, 2020.
79. "2016 massry-prize-winners" (https://hscnews.usc.edu/massry-prize-winners-to-speak-at-campus-event/). October 4, 2016. "karl-deisseroth-wins-2016-massry-prize-for-optogenetics-work" (http://med.stanford.edu/news/all-news/2016/08/karl-deisseroth-wins-2016-massry-prize-for-optogenetics-work.html). August 15, 2016. Archived (https://web.archive.org/web/20191212022550/http://med.stanford.edu/news/all-news/2016/08/karl-deisseroth-wins-2016-massry-prize-for-optogenetics-work.html) from the original on December 12, 2019. Retrieved April 2, 2020.
80. "2017 Redelsheimer Award" (https://web.archive.org/web/20190916213756/https://www.eurekalert.org/pub_releases/2017-05/pmg-kdr042817.php). Archived from the original (https://www.eurekalert.org/pub_releases/2017-05/pmg-kdr042817.php) on September 16, 2019. Retrieved April 2, 2020.
81. "2017 Fresenius Prize" (https://www.ekfs.de/en/scientific-funding/prize-for-medical-research/else-kroener-fresenius-prize-for-medical-research-2017). "Fresenius" (https://web.stanford.edu/group/dlab/media/documents/fresenius.pdf) (PDF). Archived (https://web.archive.org/web/20200310014827/https://web.stanford.edu/group/dlab/media/documents/fresenius.pdf) (PDF) from the original on March 10, 2020. Retrieved April 2, 2020.
82. "2017 Harvey Prize" (https://web.stanford.edu/group/dlab/media/documents/harveylecture.pdf) (PDF). Archived (https://web.archive.org/web/20200310014825/https://web.stanford.edu/group/dlab/media/documents/harveylecture.pdf) (PDF) from the original on March 10, 2020. Retrieved April 2, 2020.
83. "2018 Leibinger karl-deisseroth" (https://gairdner.org/award_winners/karl-deisseroth/).
84. "2018 Eisenberg" (https://members.depressioncenter.org/external/news/article/?articleId=2584).
85. "2018 Gairdner karl-deisseroth" (https://gairdner.org/award_winners/karl-deisseroth/).
86. "Kyoto Prize, Inamori Foundation" (http://www.kyotoprize.org/en/). *Kyoto Prize, Inamori Foundation*. Archived (https://web.archive.org/web/20191012071611/https://www.kyotoprize.org/en/) from the original on October 12, 2019. Retrieved March 13, 2019. "karl-deisseroth-wins-kyoto-prize-for-optogenetics.html" (https://med.stanford.edu/news/all-news/2018/06/karl-deisseroth-wins-kyoto-prize-for-optogenetics.html). May 31, 2017. Archived (https://web.archive.org/web/20200625065948/https://med.stanford.edu/news/all-news/2018/06/karl-deisseroth-wins-kyoto-prize-for-optogenetics.html) from the original on June 25, 2020. Retrieved April 2, 2020.
87. Hofschneider, Mark. "Light-sensitive microbial proteins and optogenetics" (https://laskerfoundation.org/winners/light-sensitive-microbial-proteins-optogenetics/). *Lasker Foundation*. Archived (https://web.archive.org/web/20221205062649/https://laskerfoundation.org/winners/light-sensitive-microbial-proteins-optogenetics/) from the original on December 5, 2022. Retrieved December 13, 2021.
88. "Karl Deisseroth, Peter Hegemann and Gero Miesenböck Awarded Horwitz Prize for Foundational Work on Optogenetics" (https://www.cuimc.columbia.edu/news/karl-deisseroth-peter-hegemann-and-gero-miesenbock-awarded-horwitz-prize-foundational-work-optogenetics). *Columbia University Irving Medical Center*. September 6, 2022. Archived (https://web.archive.org/web/20260225221215/https://www.cuimc.columbia.edu/news/karl-deisseroth-peter-hegemann-and-gero-miesenbock-awarded-horwitz-prize-foundational-work-optogenetics) from the original on February 25, 2026. Retrieved October 5, 2026.
89. "The Japan Prize Foundation" (https://www.japanprize.jp/). *The Japan Prize Foundation*. Archived (https://web.archive.org/web/20230902010400/https://www.japanprize.jp/) from the original on September 2, 2023. Retrieved January 24, 2023.
90. 이지현 (March 18, 2025). "칼 다이서로스 美 스탠포드대 교수 등 아산의학상 수상" (https://www.edaily.co.kr/News/Read?newsId=03476806642104632&mediaCodeNo=257&OutLnkChk=Y) [Professor Karl Deisseroth of Stanford University and others receive the Asan Medical Award]. *이데일리* (in Korean). Retrieved November 4, 2025.
91. "칼 다이서로스·안명주 아산의학상" (https://www.chosun.com/national/welfare-medical/2025/01/22/CDVVH6IT7RD53JCMGRB6A5C7WQ/?utm_source=naver&utm_medium=referral&utm_campaign=naver-news) [Karl Deisseroth and Ahn Myung-joo receive Asan Medical Award]. *The Chosun Ilbo* (in Korean). January 22, 2025. Retrieved November 4, 2025.

External links

- Karl Deisseroth (https://www.nobelprize.org/laureate/1061) on Nobelprize.org
- Official website (http://deisseroth.com)
- Academic homepage (http://www.stanford.edu/group/dlab/about_pi.html)
- Howard Hughes Medical Institute bio (http://www.hhmi.org/research/ecs/deisseroth_bio.html) Archived (https://web.archive.org/web/20130423173832/http://www.hhmi.org/research/ecs/deisseroth_bio.html) April 23, 2013, at the Wayback Machine
- Stanford OTL Inventor Portfolio – Karl Deisseroth (http://techfinder.stanford.edu/technology_listing.php?search=innovator&ID=30642)

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